```
Enter password: ********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.29 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use university_164;
Database changed

mysql> INSERT INTO department
    -> (dept_id, dept_name, email_id, phone_no, intake, staff_id)
    -> VALUES
    -> (101, 'COMPS', 'comps@univ164.edu', 9820020101, 120, 1);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`university_164`.`department`, CONSTRAINT `fk_dept_staff` FOREIGN KEY (`staff_id`) REFER
ENCES `staff` (`staff_id`) ON DELETE SET NULL ON UPDATE CASCADE)

mysql> INSERT INTO staff
    -> (staff_id, f_name, l_name, address, qualification, doj, ph_no)
    -> VALUES
    -> (11, 'Test', 'User', 'Thane', 'B.Sc', '2023-01-10', NULL);
ERROR 1048 (23000): Column 'ph_no' cannot be null

mysql> INSERT INTO staff
    -> (staff_id, f_name, l_name, address, qualification, doj, ph_no)
    -> VALUES
    -> (1, 'Anil', 'Kulkarni', 'Thane', 'PhD', '2015-06-15', 9820011001),
    -> (2, 'Meera', 'Iyer', 'Dadar', 'M.Tech', '2018-07-02', 9820011002),
    -> (3, 'Rajesh', 'Deshmukh', 'Borivali', 'M.Tech', '2017-01-20', 9820011003),
    -> (4, 'Sunita', 'Patil', 'Vile Parle', 'PhD', '2012-08-10', 9820011004),
    -> (5, 'Vikram', 'Joshi', 'Andheri', 'M.Tech', '2019-09-01', 9820011005),
    -> (6, 'Neha', 'Bhatt', 'Bandra', 'B.Sc', '2020-02-17', 9820011006),
    -> (7, 'Amit', 'Chavan', 'Kurla', 'MBA', '2016-11-05', 9820011007),
    -> (8, 'Pooja', 'Naik', 'Malad', 'B.Com', '2021-03-22', 9820011008),
    -> (9, 'Sandeep', 'Rane', 'Ghatkopar', '12th', '2014-04-14', 9820011009),
    -> (10, 'Kavita', 'Shetty', 'Juhu', 'B.A.', '2022-06-30', 9820011010);
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> INSERT INTO teaching
    -> (staff_id, specialization, no_of_pub)
    -> VALUES
    -> (1, 'Databases', 14),
    -> (2, 'Networking', 9),
    -> (3, 'Machine Learning', 17),
    -> (4, 'Signal Proc', 11),
    -> (5, 'Thermodynam', 6);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO non_teaching
    -> (staff_id, grade, skills)
    -> VALUES
```

```
    -> (6, 'A1', 'Laboratory'),
    -> (7, 'B1', 'Accounts'),
    -> (8, 'B2', 'Library'),
    -> (9, 'C1', 'Clerical'),
    -> (10, 'C2', 'Maintenance');
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO department
    -> (dept_id, dept_name, email_id, phone_no, intake, staff_id)
    -> VALUES
    -> (101, 'COMPS', 'comps@univ164.edu', 9820020101, 120, 1),
    -> (102, 'IT', 'it@univ164.edu', 9820020102, 60, 2),
    -> (103, 'CSE-DS', 'csds@univ164.edu', 9820020103, 60, 3),
    -> (104, 'EXTC', 'extc@univ164.edu', 9820020104, 60, 4),
    -> (105, 'MECH', 'mech@univ164.edu', 9820020105, 60, 5),
    -> (106, 'CIVIL', 'civil@univ164.edu', 9820020106, 60, NULL);
Query OK, 6 rows affected (0.04 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> INSERT INTO student
    -> (sap_id, f_name, l_name, address, mobile, cet)
    -> VALUES
    -> (60004250164, 'Swanandi Mahesh', 'More', 'Borivali', 9876500164, 98.52);
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO student
    -> (sap_id, f_name, l_name, address, mobile, cet)
    -> VALUES
    -> (60004250164, 'Tamanna', 'Rupapra', 'Mulund', 9876500165, 97.80);
ERROR 1062 (23000): Duplicate entry '60004250164' for key 'student.PRIMARY'

mysql> INSERT INTO student
    -> (sap_id, f_name, l_name, address, mobile, cet)
    -> VALUES
    -> (60004250170, 'Test', 'User', 'Thane', 9876500170, 101.50);
ERROR 3819 (HY000): Check constraint 'chk_cet_percentile' is violated.

mysql> INSERT INTO student
    -> (sap_id, f_name, l_name, address, mobile, cet)
    -> VALUES
    -> (60004250165, 'Tamanna', 'Rupapra', 'Mulund', 9876500165, 97.80),
    -> (60004250166, 'Sobia', 'Khan', 'Bandra', 9876500166, 96.45),
    -> (60004250167, 'Kapil', 'Nandreka', 'Ghatkopar', 9876500167, 95.90),
    -> (60004250168, 'Daksh', 'Mistry', 'Andheri', 9876500168, 98.10),
    -> (60004250169, 'Sania', 'Phatak', 'Dadar', 9876500169, 97.25);
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO course
    -> (c_code, c_name, type, de_id, s_id)
    -> VALUES
    -> ('CS101', 'Data Structures', 'THEORY', 101, 1),
    -> ('CS102', 'Database Systems', 'THEORY', 101, 1),
    -> ('IT201', 'Computer Networks', 'THEORY', 102, 2),
    -> ('DS301', 'Machine Learning', 'THEORY', 103, 3),
    -> ('EC201', 'Signals and Systems for Communication Engineering', 'THEORY', 104, 4),
    -> ('ME101', 'Thermodynamics', 'THEORY', 105, 5);
ERROR 1406 (22001): Data too long for column 'c_name' at row 5
```

```
mysql> INSERT INTO course
    -> (c_code, c_name, type, de_id, s_id)
    -> VALUES
    -> ('CS101', 'Data Structures', 'THEORY', 101, 1),
    -> ('CS102', 'Database Systems', 'THEORY', 101, 1),
    -> ('IT201', 'Computer Networks', 'THEORY', 102, 2),
    -> ('DS301', 'Machine Learning', 'THEORY', 103, 3),
    -> ('EC201', 'Signals and Systems', 'THEORY', 104, 4),
    -> ('ME101', 'Thermodynamics', 'THEORY', 105, 5);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> INSERT INTO st_email_id VALUES
    -> (60004250164, swanandi.more@gmail.com);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '@gmail.com)' at lin
e 2

mysql> INSERT INTO st_email_id
    -> (sap_id, email_id)
    -> VALUES
    -> (60004250164, 'swanandi.more@gmail.com'),
    -> (60004250164, 'swanandi.more@univ164.edu'),
    -> (60004250165, 'tamanna.r@gmail.com'),
    -> (60004250166, 'sobia.khan@gmail.com'),
    -> (60004250167, 'kapil.n@gmail.com'),
    -> (60004250168, 'daksh.mistry@gmail.com'),
    -> (60004250169, 'sania.phatak@gmail.com');
Query OK, 7 rows affected (0.02 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> INSERT INTO dependent
    -> (staff_id, d_name, relation)
    -> VALUES
    -> (1, 'Rohit', 'Son'),
    -> (1, 'Anjali', 'Wife'),
    -> (3, 'Ishita', 'Daughter'),
    -> (5, 'Mahesh', 'Father'),
    -> (7, 'Rekha', 'Wife'),
    -> (9, 'Aarav', 'Son');
Query OK, 6 rows affected (0.04 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM staff;
+----------+---------+----------+------------+---------------+------------+------------+
| staff_id | f_name  | l_name   | address    | qualification | doj        | ph_no      |
+----------+---------+----------+------------+---------------+------------+------------+
|        1 | Anil    | Kulkarni | Thane      | PhD           | 2015-06-15 | 9820011001 |
|        2 | Meera   | Iyer     | Dadar      | M.Tech        | 2018-07-02 | 9820011002 |
|        3 | Rajesh  | Deshmukh | Borivali   | M.Tech        | 2017-01-20 | 9820011003 |
|        4 | Sunita  | Patil    | Vile Parle | PhD           | 2012-08-10 | 9820011004 |
|        5 | Vikram  | Joshi    | Andheri    | M.Tech        | 2019-09-01 | 9820011005 |
|        6 | Neha    | Bhatt    | Bandra     | B.Sc          | 2020-02-17 | 9820011006 |
|        7 | Amit    | Chavan   | Kurla      | MBA           | 2016-11-05 | 9820011007 |
|        8 | Pooja   | Naik     | Malad      | B.Com         | 2021-03-22 | 9820011008 |
|        9 | Sandeep | Rane     | Ghatkopar  | 12th          | 2014-04-14 | 9820011009 |
|       10 | Kavita  | Shetty   | Juhu       | B.A.          | 2022-06-30 | 9820011010 |
+----------+---------+----------+------------+---------------+------------+------------+
10 rows in set (0.02 sec)
```

```
mysql> SELECT * FROM teaching;
+----------+------------------+-----------+
| staff_id | specialization   | no_of_pub |
+----------+------------------+-----------+
|        1 | Databases        |        14 |
|        2 | Networking       |         9 |
|        3 | Machine Learning |        17 |
|        4 | Signal Proc      |        11 |
|        5 | Thermodynam      |         6 |
+----------+------------------+-----------+
5 rows in set (0.03 sec)

mysql> SELECT * FROM non_teaching;
+----------+-------+-------------+
| staff_id | grade | skills      |
+----------+-------+-------------+
|        6 | A1    | Laboratory  |
|        7 | B1    | Accounts    |
|        8 | B2    | Library     |
|        9 | C1    | Clerical    |
|       10 | C2    | Maintenance |
+----------+-------+-------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM department;
+---------+-----------+------------------+------------+--------+----------+
| dept_id | dept_name | email_id         | phone_no   | intake | staff_id |
+---------+-----------+------------------+------------+--------+----------+
|     101 | COMPS     | comps@univ164.edu | 9820020101 |    120 |        1 |
|     102 | IT        | it@univ164.edu    | 9820020102 |     60 |        2 |
|     103 | CSE-DS    | csds@univ164.edu  | 9820020103 |     60 |        3 |
|     104 | EXTC      | extc@univ164.edu  | 9820020104 |     60 |        4 |
|     105 | MECH      | mech@univ164.edu  | 9820020105 |     60 |        5 |
|     106 | CIVIL     | civil@univ164.edu | 9820020106 |     60 |     NULL |
+---------+-----------+------------------+------------+--------+----------+
6 rows in set (0.01 sec)

mysql> SELECT * FROM student;
+-------------+-----------------+----------+-----------+------------+-------+
| sap_id      | f_name          | l_name   | address   | mobile     | cet   |
+-------------+-----------------+----------+-----------+------------+-------+
| 60004250164 | Swanandi Mahesh | More     | Borivali  | 9876500164 | 98.52 |
| 60004250165 | Tamanna         | Rupapra  | Mulund    | 9876500165 | 97.80 |
| 60004250166 | Sobia           | Khan     | Bandra    | 9876500166 | 96.45 |
| 60004250167 | Kapil           | Nandreka | Ghatkopar | 9876500167 | 95.90 |
| 60004250168 | Daksh           | Mistry   | Andheri   | 9876500168 | 98.10 |
| 60004250169 | Sania           | Phatak   | Dadar     | 9876500169 | 97.25 |
+-------------+-----------------+----------+-----------+------------+-------+
6 rows in set (0.01 sec)

mysql> SELECT * FROM course;
+--------+--------------------+-----------+-------+------+
| c_code | c_name             | type      | de_id | s_id |
+--------+--------------------+-----------+-------+------+
| CS101  | Data Structures    | THEORY    |   101 |    1 |
| CS102  | DBMS               | PRACTICAL |   101 |    1 |
| DS301  | Machine Learning   | THEORY    |   103 |    3 |
| EC201  | Signals and Systems | THEORY    |   104 |    4 |
```

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| IT201  | Computer Networks   | THEORY    |   102 |    2 |
| ME101  | Thermodynamics      | THEORY    |   105 |    5 |
+--------+---------------------+-----------+-------+------+
6 rows in set (0.02 sec)

mysql> SELECT * FROM st_email_id;
+-------------+--------------------------+
| sap_id      | email_id                 |
+-------------+--------------------------+
| 60004250164 | swanandi.more@gmail.com  |
| 60004250164 | swanandi.more@univ164.edu |
| 60004250165 | tamanna.r@gmail.com      |
| 60004250166 | sobia.khan@gmail.com     |
| 60004250167 | kapil.n@gmail.com        |
| 60004250168 | daksh.mistry@gmail.com   |
| 60004250169 | sania.phatak@gmail.com   |
+-------------+--------------------------+
7 rows in set (0.02 sec)

mysql> SELECT * FROM dependent;
+----------+--------+----------+
| staff_id | d_name | relation |
+----------+--------+----------+
|        1 | Anjali | Wife     |
|        1 | Rohit  | Son      |
|        3 | Ishita | Daughter |
|        5 | Mahesh | Father   |
|        7 | Rekha  | Wife     |
|        9 | Aarav  | Son      |
+----------+--------+----------+
6 rows in set (0.00 sec)

mysql> INSERT INTO staff
    -> (staff_id, f_name, l_name, address, qualification, doj, ph_no)
    -> VALUES
    -> (11, 'Girish', 'Pawar', 'Sion', 'M.Tech', '2023-08-01', 9820011011);
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO department
    -> (dept_id, dept_name, email_id, phone_no, intake, staff_id)
    -> VALUES
    -> (107, 'CHEM', 'chem@univ164.edu', 9820020107, 40, 11);
Query OK, 1 row affected (0.02 sec)

mysql> SELECT dept_id, dept_name, staff_id FROM department WHERE dept_id = 107;
+---------+-----------+----------+
| dept_id | dept_name | staff_id |
+---------+-----------+----------+
|     107 | CHEM      |       11 |
+---------+-----------+----------+
1 row in set (0.03 sec)

mysql> UPDATE staff
    -> SET staff_id = 12
    -> WHERE staff_id = 11;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT dept_id, dept_name, staff_id FROM department WHERE dept_id = 107;
```

```
+---------+-----------+----------+
| dept_id | dept_name | staff_id |
+---------+-----------+----------+
|     107 | CHEM      |       12 |
+---------+-----------+----------+
1 row in set (0.00 sec)

mysql> DELETE FROM staff
    -> WHERE staff_id = 12;
Query OK, 1 row affected (0.03 sec)

mysql> SELECT dept_id, dept_name, staff_id FROM department WHERE dept_id = 107;
+---------+-----------+----------+
| dept_id | dept_name | staff_id |
+---------+-----------+----------+
|     107 | CHEM      |     NULL |
+---------+-----------+----------+
1 row in set (0.00 sec)

mysql> DELETE FROM department
    -> WHERE dept_id = 107;
Query OK, 1 row affected (0.03 sec)

mysql> UPDATE COURSE
    -> SET C_NAME = 'DBMS',
    -> TYPE = 'PRACTICAL'
    -> WHERE C_CODE = 'CS102';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM COURSE;
+--------+--------------------+-----------+-------+------+
| c_code | c_name             | type      | de_id | s_id |
+--------+--------------------+-----------+-------+------+
| CS101  | Data Structures    | THEORY    |   101 |    1 |
| CS102  | DBMS               | PRACTICAL |   101 |    1 |
| DS301  | Machine Learning   | THEORY    |   103 |    3 |
| EC201  | Signals and Systems | THEORY    |   104 |    4 |
| IT201  | Computer Networks  | THEORY    |   102 |    2 |
| ME101  | Thermodynamics     | THEORY    |   105 |    5 |
+--------+--------------------+-----------+-------+------+
6 rows in set (0.00 sec)

mysql> INSERT INTO course
    -> (c_code, c_name, type, de_id, s_id)
    -> VALUES
    -> ('ME102', 'Fluid Mechanics', 'THEORY', 105, 5);
Query OK, 1 row affected (0.04 sec)

mysql> DELETE FROM COURSE
    -> WHERE C_CODE = 'ME102';
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM COURSE;
+--------+--------------------+-----------+-------+------+
| c_code | c_name             | type      | de_id | s_id |
+--------+--------------------+-----------+-------+------+
| CS101  | Data Structures    | THEORY    |   101 |    1 |
| CS102  | DBMS               | PRACTICAL |   101 |    1 |
```

```
| DS301  | Machine Learning    | THEORY    |   103 |    3 |
| EC201  | Signals and Systems | THEORY    |   104 |    4 |
| IT201  | Computer Networks   | THEORY    |   102 |    2 |
| ME101  | Thermodynamics      | THEORY    |   105 |    5 |
+--------+---------------------+-----------+-------+------+
6 rows in set (0.01 sec)

mysql> TRUNCATE teaching;
ERROR 1701 (42000): Cannot truncate a table referenced in a foreign key constraint (`university_164`.`course`, CONSTRAINT `fk_course_teaching`)

mysql> TRUNCATE st_email_id;
Query OK, 0 rows affected (0.20 sec)

mysql> SELECT * FROM st_email_id;
Empty set (0.01 sec)

mysql> INSERT INTO st_email_id
    -> (sap_id, email_id)
    -> VALUES
    -> (60004250164, 'swanandi.more@gmail.com'),
    -> (60004250164, 'swanandi.more@univ164.edu'),
    -> (60004250165, 'tamanna.r@gmail.com'),
    -> (60004250166, 'sobia.khan@gmail.com'),
    -> (60004250167, 'kapil.n@gmail.com'),
    -> (60004250168, 'daksh.mistry@gmail.com'),
    -> (60004250169, 'sania.phatak@gmail.com');
Query OK, 7 rows affected (0.02 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> SELECT COUNT(*) AS total_emails FROM st_email_id;
+--------------+
| total_emails |
+--------------+
|            7 |
+--------------+
1 row in set (0.01 sec)

mysql> exit
Bye
```